

Hotel Booking Analytics: Customer Retention & Dynamic Pricing Analysis

Project Overview

This project analyzes hotel booking data to identify cancellation patterns, customer behavior, seasonal demand, and pricing trends using SQL, Python, Machine Learning, and Power BI.

Week 1

Data cleaning, missing value handling, outlier checks, duplicate removal, and feature engineering.

Week 2

Exploratory Data Analysis including distributions, correlation analysis, cancellation analysis, and seasonal demand trends.

Week 3

Predictive modeling using Logistic Regression and Decision Tree models with evaluation metrics such as Accuracy and ROC-AUC.

Week 4

Power BI dashboard development including KPI cards, customer segmentation, cancellation analysis, and pricing dashboards.

SQL Work

Performed data import, aggregation, filtering, grouping, segmentation, revenue analysis, and advanced SQL querying in PostgreSQL.

Business Insights

Lead time, deposit type, and previous cancellations were key factors influencing booking cancellations. Seasonal demand impacted ADR and booking volume.

Conclusion

The project demonstrates an end-to-end analytics workflow suitable for hospitality business decision-making.

TOOLS USED:

Tools & Technologies: I USED PostgreSQL, Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn), Power BI, Jupyter Notebook, GitHub.